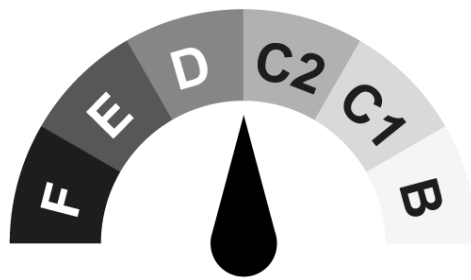




Nätverket för digital kvalitetssäkring

Competence assurance degrees



**Kvalitetssäkrad
Kompetens**

Quality assurance of competence in different competence assurance degrees is included as part of the competence model developed within the framework of the "Quality-Assured Competence" program and is managed by Kvalitetssäkringsrådet (the NDQ network).

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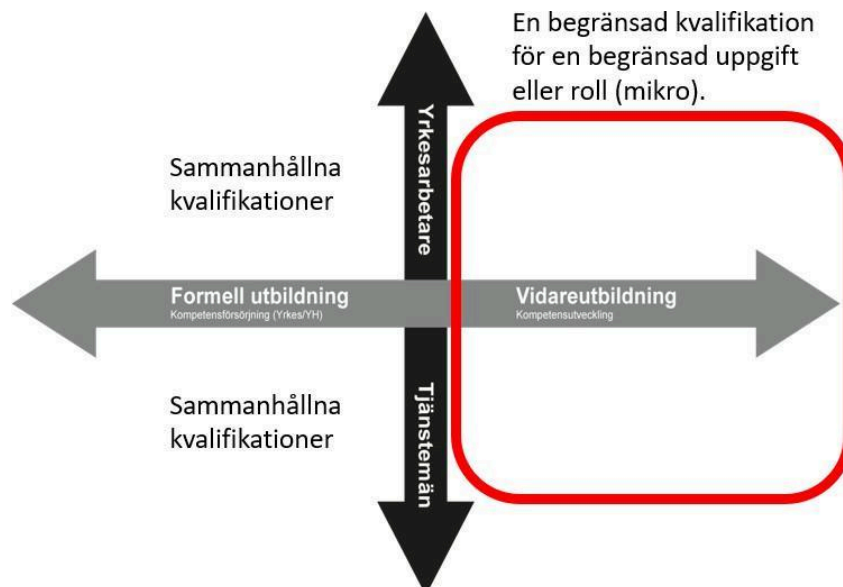
Background

The competence assurance degrees are developed by NDQ, "Nätverket för digital kompetenssäkring".

The purpose of the network is to exchange experiences, compare needs and find common and cost-effective solutions. To create collaborations, pilot projects, standardize working methods and inform important affected groups about these in order to raise both the level of knowledge and the quality of competence assessments and competence-enhancing efforts.

The competence assurance degrees are an initiative to summarise the material available and the experiences from different actors in a simple and clear way. It is intended to be used as information material, a checklist and a basis for the procurement/purchase of competence efforts within the field.

The competence assurance degrees are applied preferentially in the field of further education. For upper secondary school, college/university or other longer education there are special regulations (Swedish National Agency for Education, UHÄ, MyH etc.) for quality assurance.



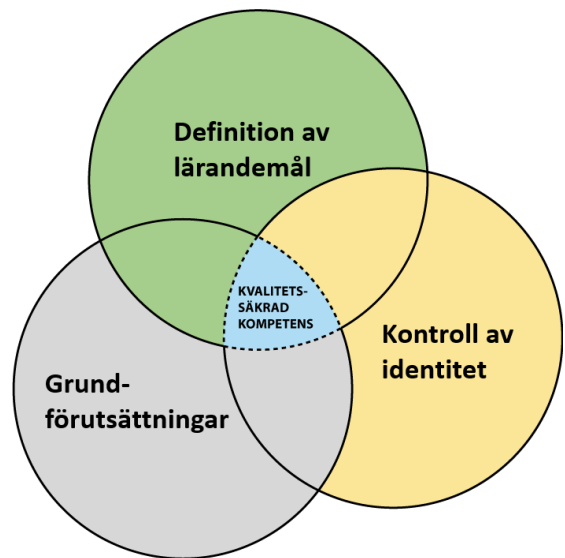
The competence assurance degrees are intended to be applicable to all sectors, industries and organizations. The majority of participants in the work come from the civil engineering sector, but the model aims to be universally valid and work in all sectors where quality assurance of competencies through competency assessments or competency-enhancing initiatives is important.

The most important standards used in the work have been:

- ISO 17024, SS-ISO 10667-2 :2020, SIS/TS 56:2023
- Individuell certifiering av kompetens, Report SIK 2014, ISBN 978-91-637-4964-3
- Datoriserad tentamen, SUNET-Inkubator, Final report 12/23/2015
- European Qualifications Framework (EQF/SeQF), EU

The competence assurance degrees are based on three main areas that all stakeholders should address.

1. **Definition of learning outcomes**, descriptions of competency according to guidelines for micro-references, EQF/SeQF. For higher levels, the competence descriptions should also be made with references.
2. **Basic prerequisites**, legitimacy of the issuer of competence, routines for competence management, e.g. attendance control, test management, how learning outcomes and tests are mapped to each other, etc.
3. **Identity verification**, what level of identity assurance is available for a certain competence assurance degree. How reliable the identity of a person with an issued certificate of competence is. Based on the standard Level of Assurance (LoA/level 1–4).¹



¹ <https://klassa.skr.se/sidor/stodmaterial/LoA>, <https://www.digg.se/digitala-tjanster/e-legitimering/tillitsnivaer-for-e-legitimering> och <https://www.elegitimation.se/>

The need for competence assurance degrees

Today there is a problem with comparing competencies. The difficulties lie in that there are no restrictions on how competencies are described. Anyone can use words such as "institute", "university", "diploma", "certification" freely. This can be misleading about the quality and weight behind the person who issued the competency. The degrees are designed to address these problems. The competence assurance degrees are largely based on already established and standardized concepts and frameworks that together enable competence assurance.

The purpose of the competence assurance degrees is to increase knowledge and awareness of the problems with comparing and evaluating competency. The purpose is also to be a support in procurements and assessment of an issued competency. We do this by standardizing and clarifying concepts and relationships, and setting a framework with descriptions and requirements for how the different grades quality-assure the competence.

Important concepts

Competence – We choose to define **competence** based on the following three standards:

Kompetensutredningen (SOU 1991:56) defines it as "the ability to cope with the different demands placed on a particular situation in a particular activity. This means that competence should always be seen in relation to a specific task or situation" (our translation). Demands in certain operations may thus include legal, environmental or industry-specific market needs.

According to the EQF, an individual's **competence** is divided into knowledge, skills, as well as responsibility and autonomy.

- **Knowledge** - Both factual knowledge and knowledge of principles and processes (information utilization through learning, theoretical or practical, facts, principles, practices). The participant should be able to...
- **Skills** - The ability to apply knowledge (apply knowledge, convert theories or practice into action/to perform tasks or explain/solve problems). Be able to use methods, calculations, tools, etc.) The participant can...
- **Responsibility and autonomy** - The ability to use different knowledge and skills together, combine different principles into new working strategies or develop working methods. The participant takes responsibility for and independently...

Traceability – The key word in ISO 17024 is "traceability". It aims to create a robust and reliable certification process where it is possible to track and verify every step of the assessment (from requirements to testing). This helps to ensure that the certification process is fair, transparent and reliable.

Learning outcomes provide direction and focus on the desired outcome of learning. They should be clearly defined, measurable, relevant and describe what students are expected to learn and achieve as a result of an educational activity. Learning outcomes are part of a **qualification description**, which further clarifies where learning outcomes come from (e.g. legal requirements, ISO standards, etc.), the outcomes are broken down into sub-goals/sub-qualifications and describes at which taxonomic level the objective is located (know – be able to apply, etc.).

Council of Qualifications – A group or body responsible for setting and overseeing a learning objective for a particular professional group or industry. These councils play an important role in ensuring that people working in a specific field have the necessary skills, knowledge and

competencies to carry out their duties effectively.

Identity verification – In this model, identity verification – ensuring reliable identity – is one of three central components. For the individual, established identity is not always crucial for benefiting from and applying knowledge in working life. However, identity verification is critical in situations such as recruitment or when a person is responsible for or performs tasks that require specific competence. The model is based on Level of Assurance (LoA 1–4) which is applied differently depending on the different competence assurance degrees.

Level 1 does not require any identification, but it is sufficient to provide, for example, a name and email address. Levels 2–4 place higher demands on the user to verify their identity.

Trust level 1: Indicates **low trust** in identity. Does not require identification, but it is sufficient to provide, for example, a name and email address.

Trust level 2: Indicates **some trust** in identity. Identity is checked through registration with a verifiable email belonging to the person combined with two-factor authentication.

Trust level 3: Indicates **high trust** in identity. Identification using a valid Swedish identification document.

Trust level 4: Indicates **very high trust** in identity. Identification using a valid Swedish identification document both for the first time and when renewing training/competence/certification.

NDQ:s Competence assurance degrees

	Sektors Certifikat (B)	Oberoende Certifikat (C1)	Certifikat (C2)	Diplom (D)	Utbildnings- intyg (E)	Information (F)	Mikromerit
Definition av lärandemål							
Grund- förutsättningar							
Kontroll av identitet							
Saknas nivå av kompetens-säkring i området	Låg nivå av kompetens-säkring i området	Mellannivå av kompetens-säkring i området	Hög nivå av kompetens-säkring i området	Hög nivå av kompetens-säkring i området	Mycket hög nivå av kompetens-säkring i området		

To explain the minimum level of quality assurance of the three components, we use “dice” from 0-4 to illustrate how advanced the requirements are for Learning outcomes, Basic prerequisites and Identity verification. Note that it is the minimum dice level that is stated, e.g. education defined as Educational Certificate (E) may have a more advanced definition of learning outcomes but still remain at Educational Certificate due to the other “dice levels” not being met.

It should be noted that a competence assurance degree (e.g. B) does not necessarily mean that the individual has learned more or better compared to another degree (e.g. F). Rather, a higher grade indicates a stronger degree of reliability and quality assurance of the competency that the individual is claimed to have through, for example, a certificate of competence.

For individuals and organizations, the needs vary depending on the situation. Critical and statutory data may be important for an organization to ensure, in which case a high level of reliability and quality assurance may be needed. In other cases, it may be very valuable for an individual to have access to a short instructional video on their smartphone “right now” that describes how to handle a task.

Verification of the individual's competence should be linked to learning outcomes but does not place any direct requirements on the forms in which learning outcomes are verified. Some different ways:

- Crediting of previous jobs, previous education, etc.
- Interview
- Oral exam
- Written exam/tests (certification, validation, etc.)
- Essay
- Workshop
- Practical exam (e.g. in a simulator or in a real facility)

The choice of methods for learning, assessment and examination forms should be linked to learning outcomes and the degree of competence assurance. For example, (F) can be achieved by including an instructional video in a training section, while (E) can include a workshop or a case study. And C2 can be tested through a written or practical exam.

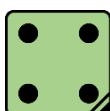
By linking learning outcomes, educational activities and assessment of learning outcomes in a curriculum map², teachers/training providers can ensure that teaching and examinations are well-structured, relevant and effective. In addition, curriculum mapping can help identify any gaps or overlaps in the education.

² **Curriculum mapping** illustrates the components and building blocks of the education. It visualizes the entire education plan and how the components of the education (goals, activities, assessments, values, etc.) relate to the overall educational goals. The mapping should answer whether we are teaching what is to be examined? Are we examining what is taught? Are we teaching what the learning objective is?

Sector certificates (B)

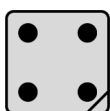
Adapted and quality assured by the industry or sector. The competences that the certificate provides have defined learning outcomes according to the SeQF standard and there is clear traceability between assessment and learning outcomes. The sector certificate should generally be able to be placed against micro-credentials at MyH (read more in Annex VI). The industry has commissioned or issues the certificate itself. Limited in time, maximum 5 years.

The certificate consists of a certified document with information about the holder's name, personal identity number, achieved learning outcomes, assessment method and issuer of the certificate.



Definition of learning outcomes

Requirements that the knowledge that forms the basis for the certification is defined in a structured and verifiable manner in a similar way as for micro-credentials or level-placed EQF/SeQF qualifications. There should be a management organization that updates it continuously.



Basic prerequisites

Requirements for clear quality procedures to ensure that the third party issuing the certificate is not the same as the one providing the training. There are procedures for how attendance is monitored, how the tests included in the training are monitored.



Identity verification

Trust level 3 should be used, physical ID verification through a nationally recognized ID document or equivalent level of e-identification.

NOTE! Given examples do not currently have to meet the method's requirements for learning objectives, basic prerequisites and identity, as well as the requirement for legitimacy as defined in this model. Our assessment is that these examples are more or less suitable for placement as Sector Certificates.

Certification for bank advisors, Aviation Certificates, Certified fiber technician (third-party certificate), Refrigeration Certificate according to category 1-5, Emergency and safety coordinator, Property manager, Security guard, etc.

Independent Certificates (third party) (C1)

The competence that the certificate provides has clearly defined learning outcomes and some traceability between assessment and learning outcomes exists. The industry has not commissioned or issues the certificate but brings together parts of the industry. There is no dependence between the certificate issuer and the trainer, i.e. certification is completely independent and carried out by a 3rd party. The trainer may not carry out the control of the learning outcomes. According to ISO 17024, the validity of the certificate is limited in time to a maximum of 5 years.

The certificate consists of a certified document with information about the holder's name, social security number, achieved learning outcomes, assessment method and issuer of the certificate.



Definition of learning outcomes

Requirements that the knowledge that forms the basis for certification is defined in a structured and verifiable manner (does not have to be but can mimic the EQF/SeQF standard). There should be an independent management organization that updates it continuously.



Basic prerequisites

Requirements for clear quality procedures to ensure that the third party issuing the certificate is not the same as the one providing the training. There are procedures for how attendance is monitored, how the tests included in the training are monitored.



Identity verification

Trust level 3 should be used, physical ID verification through a nationally recognized ID document or equivalent level of e-identification.

NOTE! Given examples do not currently have to meet the method's requirements for learning objectives, basic prerequisites and identity, as well as the requirement for legitimacy as defined in this model. Our assessment is that these examples are more or less suitable for placement as Independent certificates.

IPMA, PMI, Authorized installer of property networks, Inspection manager according to BBR, Certified fire protection expert, etc.

Certificates (other party) (C2)

The competence of a certificate has defined learning outcomes. Assessment and learning outcomes must match each other but do not need to be traceable. The certificate can be initiated by interest organizations or a company that can also provide training itself, there may be a dependency between the certificate and the training provider. According to ISO 17024, the validity of the certificate is limited in time to a maximum of 5 years. The trainer may not carry out the control of the learning outcomes.

The certificate consists of a certified document with information about the holder's name, personal identity number, achieved learning outcomes, assessment method and issuer of the certificate.



Definition of learning outcomes

Requirement that the knowledge that forms the basis for the certification is defined in a structured and verifiable manner.



Basic prerequisites

Requirements for clear quality procedures to ensure that the instructor cannot influence the results of completed competency tests. There are procedures for how attendance is checked and how the tests included in the training are checked.



Identity verification

Trust level 3 should be used, physical ID verification through a nationally recognized ID document or equivalent level of e-identification.

NOTE! Given examples do not currently have to meet the method's requirements for learning objectives, basic prerequisites and identity, as well as the requirement for legitimacy as defined in this model. Our assessment is that these examples are more or less suitable for placement as Certificates (other party).

Flammable hot work, Certification for wet room work (e.g. BKR/GVK), Microsoft Azure certification, Construction Industry Organizer Training, Safe water, etc.

Diploma (D)

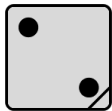
Requirement for some form of control over whether the participant has actually acquired the knowledge and/or skills that the training aimed to achieve in order to pass the certification. This control can be carried out by the training company itself and can consist of, for example, knowledge tests, practical exercises with supervision or project work.

The certificate consists of a certified document with information about the holder's name, personal identification number (if applicable), achieved learning outcomes, assessment method and issuer of the certificate.



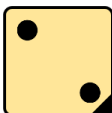
Definition of learning outcomes

Requirements that the knowledge that forms the basis for certification is defined in a structured manner.



Basic prerequisites

There are quality procedures to ensure the content of the training. There are procedures for how attendance is monitored and how the tests included in the training are monitored.



Identity verification

If a social security number is applicable for the diploma, trust level 3 applies, physical ID check through a nationally recognized ID document or the equivalent level for e-identification. If a social security number is not applicable, trust level 2 applies, such as registration with a verifiable email or registration with e.g. Bank-ID or Freja-ID.

Most courses/educations could be classified as Diploma (D). By ensuring clear learning objectives and relevant knowledge verification and quality procedures, this is a goal for regular educations. Diploma (D) is covered by training with mixed forms of training over a longer calendar period, as well as ongoing knowledge tests and practical cases.

Education certificate (E)

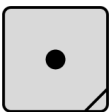
Limited verification that the participant has attended and taken part in the training content, e.g. in the form of a requirement to participate in most of the training.

The certificate consists of a document with information about the holder's name and the issuer of the certificate.



Definition of learning outcomes

Available in an overall form, e.g. a course description where the purpose and content of the training is defined.



Basic prerequisites

There are quality procedures to ensure attendance and participation in the training.



Identity verification

Trust level 1, no identification is required, only name and email address.

Educations that have a lower need for quality assurance. For example, online courses, corporate training, specific courses in, for example, presentation techniques, etc. Courses that are aimed at enabling the individual to perform the job in a better way but where verification is not important.

Information (Just in time learning) (F)

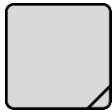
Information/conferences/educations/courses/instructional videos that lack verification of the participant's identity, control over their presence and that have a general course description. For example, MOOC - (Massive Open Online Courses) or lecture/information session. This type of learning often takes place on the individual's initiative, a good way of learning in working life.

No proof is issued.



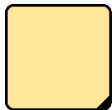
Definition of learning outcomes

No requirement for defined learning outcomes.



Basic prerequisites

No requirement for quality routines.



Identity verification

No legitimation required, often no registration required at all.

Educations that have an internal or limited need to demonstrate quality assurance. The content of the training may be well-developed, but competence assurance according to this model's standard is a low priority. For example, onboarding for new employees, etc. Also open online training that may occur at e.g. the Swedish National Board of Housing, Building and Planning, the Swedish Work Environment Authority, the Swedish Civil Contingencies Agency (MSB), etc. Information sessions/webinars or instructional videos on YouTube etc., from an organization or authority.

Overview of content/requirements for various components

The table briefly describes what needs to be met at the different competence assurance degrees.

Learning outcomes	B	C 1	C 2	D	E	F
- Learning outcomes according to EQF or equivalent broken down into sub-areas/sub-qualifications	X	X	X			
- Learning outcomes				X		
- Purpose/content					X	
- No detailed description/information (unclear content)						X
Basic prerequisite for test and assessment	B	C 1	C 2	D	E	F
- Traceable against formulated learning outcomes and targets according to EQF or equivalent	X	X	X			
- Set against formulated learning outcomes				X		
- For digital exams: randomized questions and answer options	X	X	X			
- Questions constructed and reviewed by knowledgeable question designers	X	X				
- Questions and practical exercises are adapted to the target group, complexity and linked to assessment matrices	X	X	X			
Basic prerequisite for legitimacy	B	C 1	C 2	D	E	F
- Certification of competence should be carried out by an independent organisation with legitimacy in the industry/sector	X	X				
Identity and attendance control (Trust level)	B	C 1	C 2	D	E	F
- Trust level 1					X	
- Trust level 2				X		
- Trust level 3	X	X	X			
- Trust level 4 (in case of recertification requirement)	X					

Appendix - Organizational and implementation requirements (Basic prerequisites)

This appendix describes how the organization at different competence assurance degrees must meet different requirements. The appendix should be seen as a support and checklist for the organization to know what is needed to assess its competence assurance degree or to increase the quality of a procurement/order.

Organizational and implementation requirements	B	C 1	C 2	D	E	F
Quality requirements for certification issuers (see I)	X	X	X			
Quality requirements for exams (see II)	X	X	X			
Quality requirements for the exam administration organization (see III)	X	X	X			
Quality requirements for different roles (see IV)	X	X	X			
- Examiner						
- Exam invigilator						

Quality requirements for certification issuers (Annex I)

The issuing organization should have a clear industry/sector connection, either through active ownership or governance and control functions. Certification issuers should also be issuers of certificates.

Organization	B	C 1	C 2	D	E	F
Legitimacy - Industry/sector stands behind the certificate	X	X				
Council for Qualifications, expert management team that oversees/is responsible for the defined learning outcomes. Should be separate from the educating organization.	X	X				
Appeal function (customer, participant)	X	X	X			
Quality policy	X	X	X			
Not only profit interest, long-term commitment	X	X				
Key stakeholders are invited to participate in Councils for Qualifications	X	X				
Roles (if applicable)	B	C 1	C 2	D	E	F
- Regarding other executives, the following or equivalent must be present (one person can hold multiple roles):						
Quality Manager - Someone who reviews/evaluates relevance/quality etc.	X	X				
Examiners, <u>approves</u> and/or assesses the individual's results	X	X	X			
Invigilator/exam administrator with the right knowledge about conducting the exam/test) - independent and impartial in relation to those/things being tested - applies to all of the above!	X	X	X			

Quality requirements for exams (Annex II)

Requirements for the examination occasion/knowledge test for the different competence assurance degrees.

Implementation	B	C 1	C 2	D	E	F
How the participant's confidentiality is ensured, e.g. consent to registration etc.	X	X	X			
Which aids are permitted and not permitted (e.g. no access to digital media, etc.)	X	X	X			
Examination process, which parts are included, maximum time, feedback on results, etc.	X	X	X			
Assessment criteria (Pass/Fail)	X	X	X			
Examination is linked to learning outcomes (Questions/tasks linked to learning outcomes)	X	X	X			
Description of Lifecycle/validity period for examination (max 5 years- ISO 17024)	X	X	X			
Description of recertification: - Prerequisites and requirements (content, completely new or just a small part, etc.) - Other requirements, e.g. medical examinations, experience, etc. - Rules and conditions for granting higher qualifications	X	X	X			
Registration and identity are secured and registered	X	X	X			
Rules for appeal, how it is handled and who decides	X	X	X			
Rules for evaluation and further development of the system	X	X	X			
Rules for keeping records of results, e.g. archiving and storage – complying with laws and regulations	X	X	X			
Routines for incident and deviation management during test execution	X	X	X			
Routines for digital distance implementation	X	X	X			

Checklist for online implementation ³	Exists	Missing	N/A
Routines for different authorization roles in IT systems for computerized/digital examinations			
Routines for ID check of the participant, e.g. analog, physical person (online exam invigilator)			
Procedures for authenticating the participant without a physical online invigilator (takes place entirely digitally)			
Routines for monitoring analogously (e.g. that unauthorized aids or other persons are not in the room)			
Routines for digital monitoring (system solution that monitors without recording via webcam, e.g. that the examinee is not replaced or has more than one person in the room)			
Routines for monitoring with screen recording and/or webcam (film recording, or taking screenshots at least x times per minute)			
Routines for plagiarism checking of text (performed text matching)			
Use of secure browser for kiosk mode that locks the use of functionality on the computer screen			
If e-meeting tools (Zoom, Teams, etc.) are used, a routine description must be available for implementation (e.g. film recording, storage)			

³ Report Datoriserad tentamen (2015) URL: [Datoriserad_Tentamen_Slutrapport2015_arkiv160107.pdf | Använder Box https://hig.app.box.com/s/rr9ltz6gkch19ra5rmm6wzdhyfi46p2r](https://hig.app.box.com/s/rr9ltz6gkch19ra5rmm6wzdhyfi46p2r)

Quality requirements for exam administration organizations (Annex III)

Examination centers that handle the test impartially (such as local Driver's Test offices, learning centers). Can be test-learning centers, education providers, employers or pure exam administration organizations.

Physical and digital implementation	B	C 1	C 2	D	E	F
Legitimate/known (organization number, address, tax ID) ⁴	X	X	X			
Procedure for impartial exam handling	X	X	X			
Documented routines and processes	X	X	X			
Routines for following instructions from demanders/orderers	X	X	X			
Procedure for access to the venue	X	X	X			
The venue is adapted to the examination format	X	X	X			

Checklist for exam administration organizations (Requirements for venues, equipment and digital environments when conducting examinations.)	Exists	Missing	N/A
Routine for incident management/deviation management (interruptions, cheating, etc.)			
Routines for breaks and bathroom breaks			
Routines for checking that toilets, waiting rooms do not have unauthorized hidden information that helps test takers to cheat, monitoring and documentation			
Routines regarding late arrival, identification, exam adaptations (certificate of documented disability), Communication – that during the exam period it is not permitted to communicate with anyone other than the exam invigilator			
Procedure for informing about evacuation routes and assembly points			
Procedure for the Personal Data Act in case of streaming surveillance			
Accessibility statement and requirements for NPF diagnoses and functional disabilities and impairments (UDL) - instructions regarding these adaptations and how assistive devices may be used.			
Industry affiliation (quality policy, e.g. Learning centers)			
Guaranteed metered and secured WiFi, broadband capacity (if needed)			
Plan and procedures for appropriate furnishing, test takers should not be able to communicate with each other.			
Routines for how outerwear and mobile phones should be placed/handed in			
Routine for updating necessary software and existing computer equipment – so that updates do not occur just before the exam time			
Proctoring – online exam monitoring, system solutions work on site with webcams/headsets			
Procedure for authentication of the exam invigilator (analogous, physical person, etc.)			
Routine for analog monitoring, physical person monitoring with e-meeting tools (e.g. ensuring that unauthorized aids or other persons are not in the room)			
Routine for digital monitoring, system solution that monitors without recording via webcam (e.g. ensuring that the test taker is not replaced or has other people in the room)			
Routine for using so-called secure browsers to lock the use of functionality on the computer screen.			
Routine for using e-meeting tools. Requirements and routines are followed and that the tool can also manage and document attendance and that submissions of tests and documentation have been made according to instructions and routines established for the digital examination.			

⁴ Certified test center, member of Nitus.se or similar. Note above about "Organization and executives" uncertainty in writing about corporate form. Also note here: What is unclear in this regard. Is it not more relevant to use the AAA Credit Rating System, via Bilnode or another existing Rating: [aaa-kreditrating-produktbroschyr- hosten-2017.pdf \(bilnode.se\)](#) or authorization via [Ansök om auktorisation – Utbildningsföretagen \(utbildningsforetagen.se\)](#)

Quality requirements for different roles (Annex IV)

Examiner

The person who is knowledgeable about and ultimately responsible for examination and assessment, requirements for impartiality. Examiner can also be called Quality Manager.

Physical and digital implementation	B	C 1	C 2	D	E	F
Requirements for competence in the subject, both theoretical and possibly practical	x	x	x			
Competence regarding learning outcomes, understanding the relationship between examination and qualification description	x	x	x			
Independent/impartial in relation to participants being tested	x	x	x			
Competence in question construction, qualitative and quantitative methods	x	x	x			

Exam invigilator

Person who starts, supervises and ends the test/examination. Should have the right knowledge about conducting the exam/test - independent and impartial in relation to those/the things being tested.

	B	C 1	C 2	D	E	F
Physical implementation						
Requirements for basic knowledge of the role	X	x	x			
Instructions established by the management team or equivalent.	x	x	X			
Independent in relation to the selection of questions if these are taken from a question bank and not correcting the test.	x	x	x			
Independent in relation to participants	x	x	x			
Instructions for exam administrators must be developed (such as table setting, equipment, aids, identification, control, handling irregularities, etc.)	x	x	x			
Knowledge of Swedish and English	x	x	x			
Virtual implementation						
In addition to the above	x	x	X			
Ability to monitor audio and video 100% in real time	x	x	X			
Access to systems/technology to monitor all participants in real time simultaneously.	X	x	x			
Knowledge and familiarity with procedures for checking identification remotely, HA or equivalent	x	x	X			

Levels of assurance (LoA) (Annex V)

LoA - Level of Assurance	Description of Trust levels - Identity verification	B	C 1	C 2	D	E	F
Trust level 1	No identification is required, it is sufficient to enter, for example, a name and email address.						X
Trust level 1	Trust level 1, no identification required, only name and email address					X	
Trust level 2	If a social security number is not applicable, Trust level 2 applies, such as registration with a verifiable email or registration with e.g. Bank-ID or Freja-ID.				X		
Trust level 3	If a social security number is applicable for the diploma, Trust level 3 applies, physical ID check through a nationally recognized ID document or the equivalent level of e-identification.				X		
Trust level 3	Trust level 3 should be used, physical ID check through nationally recognized ID document or equivalent level of e-identification.	X	X	X			
Trust level 3	Trust level with digital ID check at the time of the exam remotely of identity via visa passport or national ID card or e.g. BankID/Freja (or equivalent) or via using recognized two-factor authentication (2FA) also called two-step verification in combination with other online monitoring functions (for example screen recording, safe browser for kiosk mode - safe exam browser, recording with webcam, biometric data, AI-stylometry and/or together with so-called online exam invigilator during the ongoing exam/test).	X	X	X	(x)		
Trust level 4	<i>Trust level 4 should be used for Recertification requirements: "The user needs to prove their identity through a personal visit with a full Swedish identification document, both the first time and when renewing every five years. The user is identified through an e-identification that is protected in a special chip, which can be on, for example, a plastic card, a mobile phone or a USB device. There is a very high level of trust in the identity, and a requirement for two-factor authentication."</i> ⁵	X					

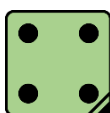
⁵ <https://www.digg.se/digitala-tjanster/e-legitimering/tillitsnivaer-for-e-legitimering#h-Svenskatillitsnivaer>

Micro-credentials (Annex VI)

A micro-credential is a recognized proof of learning outcomes that a person has achieved over a short period of time, according to transparent standards and requirements and after an assessment. Micro-credentials are an EU standard that has not yet been introduced in Sweden. The Swedish National Agency for Vocational Education and Training (MyH) has been appointed by the government to introduce micro-credentials in Sweden. MyH should level micro-credentials in the future but has not yet started doing so. An individual and shorter course at a university or college is at the same level as micro-credentials as they have a corresponding system that follows the “Bologna process”⁶

A level placement of MyH, unlike Sector certificates, does not necessarily mean that the sector/industry has approved the micro-credential in the same way as with a Sector certificate in this model. MyH's models lack sufficient requirements for Basic prerequisites and Identity verification that this model describes and thus do not ensure comprehensive legitimacy in the sector.

Micro-credentials belong to the individual. They can be shared, transferred and included in more extensive credentials or qualifications. (www.myh.se)



Definition of learning outcomes

Requirement that the knowledge that underlies the certification is defined in a structured and verifiable manner in accordance with micro-credentials or EQF/SeQF. To fully meet the “requirements”, level placement with MyH is required. However, there is currently no guarantee that MyH checks that the sector is behind it, so the value of this micro-credential will not be as great and will not necessarily be approved in the same way as with a Sector certificate (B)



Basic prerequisites

NOTE! There are no requirements from MyH linked to the Basic prerequisites as described in this model. The competence assurance degrees set requirements beyond the Definition of learning outcomes, which micro-credentials and level placement of MyH do not. In comparison with Sector Certificates, there are no requirements for clear quality routines to ensure that the third party issuing the certificate is not the same as the one providing the training. There are also no requirements for the other Basic prerequisites that this model sets out.



Identity verification

NOTE: There are no requirements from MyH related to Identity verification.

⁶ <https://education.ec.europa.eu/education-levels/higher-education/inclusive-and-connected-higher-education/european-credit-transfer-and-accumulation-system>